ActInf Livestream #016.1: "Neural correlates of consciousness under the F

Livestream | 1:52:31

hello
everyone thank you very much for joining
this is the active inference lab
and you are listening to the active
inference live stream welcome to the
active inference lab
everyone we are an experiment in online
team communication
learning and practice related to active
inference you can find us
on our website twitter email
youtube discord or key base team this is
a recorded and an archived live stream
so
please provide us with feedback so that
we can improve on our work
all backgrounds and perspectives are
welcome here
and we will follow good conversational
live stream etiquette
today we're really excited to be
discussing
the neural correlates of consciousness
under the free energy principle
with one of the two authors juan juan
jolisa
and um today we're going to be having
our first discussion
16.1 and then next week we'll be having
a follow-up discussion on the same paper
so if you have more thoughts or
questions arising in the next week
that will be a great time to bring them
up
today in act stream 16.1

our goal is to be learning discussing
and
hearing from everyone and that is going
to be facilitated by
this awesome presentation that we are
about to
hear from wanja so for 16.1
we're gonna have a presentation um
and then we're gonna just pick up with
the participatory discussion
so laundry thanks again for joining us
and please take it away
thank you for having me it's a real
pleasure to
have the opportunity to present this
paper here and discuss it with you
i'm just gonna share my slides
okay i think you can see the looks great
um
first i'd like to
make a short um
advertisement for my journal so together
with
sasha fing and jennifer wind i founded
philosophy in the mind sciences
an open access journal for
work um at the intersection of
philosophy neuroscience and psychology
and we recently published
a special issue on the neural correlates
of consciousness
which you can find at
philosophymindscience.org so if you're
interested in this topic you might want
to
check this special issue out
now in this short presentation
i'm going to mention just some some key

points from the
preprint and this is the structure
of the paper these are just section
titles
after some more sections after the
introduction
and aren't we revising the paper so we
received some
very useful reviewer comments
and actually one of the rarities
pointed out that if you're not already
committed to or interested
in the free energy principle
the results that we present in this
paper may not be that
interesting or not may not see that
seeing that
relevant and in fact and in part this is
i think a fair comment because in the
first
section after the introduction we
review the standard notion of a neural
correlate of consciousness
which has been provided by david
chalmers in a seminal paper
in 2000 and we then mentioned some
challenges for this concept
and this motivates moving from
the neural corners of consciousness to
the computational correlates
and um actually then in the next
discussion we also
relate this to the free energy principle
and then it turns out that there are
some problems if you want to apply
the free energy principle to this
kind of research because the main reason
is that the free energy principle

is not itself a theory of consciousness
and um it's it's not a
theory of computational correlation of
consciousness so
um there are first some obstacles that
have to be overcome
if you want to apply the free energy
principle to research on consciousness
or
on neural and computational correlates
of consciousness
in particular and if you're not already
committed to
the free energy principle or not not
really interested in it
um you may wonder well why should i even
bother thinking about this if it only
creates new problems
that you don't get into if you're not
dealing with the free energy principle
in the first place so
um this is i think a fair question and
um we we're trying to highlight
the um genuine contributions that we're
making or that we're trying to make in
this paper
and the revisions and these
mainly relate to the final two sections
before the conclusion
in which we talk about computational
explanations of consciousness in general
and what
the free energy principle can provide or
can contribute to this
this project and um then we
try to try to explain what the
specific role of the free energy
principle can be here and we suggest

that it may provide
unification so i will
um take a few minutes to talk about
these things in a bit more detail
um let's see this is a
more detailed um overview of the
the paper so in the
section in which we talk about general
currents of consciousness we
mention that according to the standard
definition a neural
correlate of consciousness is a
minimally sufficient neural
structure and then there are three
challenges that we mentioned one is that
actually what is interesting about the
neural activity
associated with consciousness may not be
that it happens in a particular place in
the brain
but more than it's a that it displaces
a certain dynamics and this can even be
a
global dynamics so um it's it's not
what's happening at the place but what's
happening over time and the
or maybe in the entire brain or in large
parts at least of the brain
then the question another question is
even if you can map certain
types of neural activity to conscious
experiences
this mapping can be seemingly arbitrary
so we
need to understand why or would like to
understand
why that particular particular type of
activity

is associated with consciousness
and then a third challenge is that
the notion of a neural correlate of
consciousness
as defined by chalmers and as used in
much research on ncc's
only applies to usual conditions
and does not apply to unusual conditions
so after brain damage for instance
and another very unusual condition
is provided by
potential cases of islands of awareness
so this is from a
recent paper by timbane and yosef and
marcelo masimini in which they
ask the question whether there can be
systems that are
causally isolated from the body and on
the environment and still be conscious
so um can there be conscious states
in systems that neither receive sensory
input
nor produce motor output
and there are some potential candidates
for such
systems which include ex-cranial brains
hemospherotomy and
cerebral organoids so
an ex-cranial brain is a brain that has
been
extracted from the cranium but is kept
alive and which can
still display some non-trivial activity
[Music]
and in hemispherotomy
a large part of the brain is
more or less disconnected from from
other parts or the rest of the brain so

neuronal connections
to other parts of the brain are cut
and in well
organoids such as neural structures
grown in the lab and in all cases
one may wonder well could such
structures
such systems give rise to activity
patterns
that are sufficient for consciousness
and what we suggest is that
computational correlates could
provide at least some progress towards
an answer
because computational correlates can be
applied to different kinds of system
and um and in fact the authors of this
paper mention
complexity measures so um
there's some evidence that neural
activity associated with consciousness
displays
a large or high complexity
high algorithmic complexity and
in principle one could measure the
complexity of activity generated in
these
these isolated systems and that could
provide some evidence
for inferring the presence of
consciousness
in such systems now um
actually um
complexity is also an issue when it
comes to applying the free energy
principle
um i will talk about that in a minute um
and one of the things we do in the next

section
section is that we relate the free
energy principle to
work on computational correlation of
consciousness and computational
principles that um were or that
have been associated with consciousness
and um we in in particular we also
discussed the challenge of islands of
awareness and this is not just a
challenge for
neural corals of consciousness but also
especially a challenge for the free
energy principle one of the problems
that
is created by trying to apply the free
energy principle
and the reason is this the free energy
principle
is mainly applicable to
systems that are in causal interaction
with their environment
so systems that are part of an action
perception loop
in which internal states interact with
external states mediated by blanket
states which
contain sensory and active states but if
you have an
isolated system with which is cut off
from sensory input and motor output
it's not clear that you can apply the
free energy principle
to describe what's going on in the
system but actually there's
there's some um existing work on
the free energy principle and dreaming
by alan hobson

and carl fristen and in
that work they suggest that well
um i just
go a bit forward um you can
as is well known i guess you can um
you you can formulate or describe the
free energy functional as a as involving
a
accuracy and a complexity term so you
can write
variation of free energy as complexity
minus accuracy or complexity plus
inaccuracy and
as it turns out the accuracy term
contains um a
term denoting particular states which
involve blanket states
and the complexity term does not so
the complexity term can be minimized by
changing
internal states and so
in previous work it was suggested that
during dreaming
the causal interaction with the
environment is attenuated and so
the this free energy gradient is mainly
driven by
the complexity term so actually so
basically what the brain's doing during
dreaming is minimizing complexity
and we suggest that the same
holds our foturi for
cases of islands of awareness so such
systems
should at least for some time
minimize free energy by minimizing
complexity
and now this brings up complexity again

and the question
how this relates to empirical research
according to which
neural activity associated with
consciousness
maximizes complexity or at least
displays
um high complexity
and um this um
well tension can be
resolved by by noting that the
um free energy functional contains
a statistical form of complexity
and complexity measures and
consciousness
signs are measures of dynamical
complexity
that are often based
on an approximation to algorithmic
complexity
so they're actually two different
notions of complexity
and play here and according to the free
energy principle islands of awareness
should
minimize one type of complexity
but it leaves open the question whether
they
maximize a different type of complexity
okay and we suggest that it would be
useful
to formulate or develop a notion
of dynamical complexity within the
framework provided by the free energy
principle
so it would be useful to define dynamic
complexity
or measure of dynamical complexity in

terms of the intrinsic
information geometry of a system as
opposed to
a definition in terms of the extrinsic
information geometry um
but this is actually not something i
want to dwell on here so
um let's go to the next point um
computational explanations
of consciousness um we
well that's a fundamental problem
associated with
trying to develop a computational
explanation of consciousness
because computations can be performed by
vastly different types of system so um
you can perform computations in the game
of life
and you you can compute mathematical
functions
by playing magic the gathering so
the game of life is during complete
and as i recently learned magic the
gathering is
turing complete as well but i wouldn't
suppose that
anything that happens in the game of
life
gives rise to consciousness or that you
can instantiate consciousness by
playing magic the gathering in a
particular way
so um just computing the right
mathematical
functions in a physical system
is not sufficient for consciousness
or in other words there's a difference
between simulating consciousness

in a system and actually instantiating
consciousness
and the question is okay how can we make
this
difference of how can we understand the
difference
between simulating and instantiating
consciousness
and here um we
think that the free energy principle can
make a
perhaps tiny but we think relevant
contribution
and the main idea is this as i already
mentioned
um the this is the causal flow in an
action perception loop um according to
the free energy principle
so here we have a physical system with
internal states um and it's
coupled to external states via
blanket states which comprise sensory
and active states and so the main causal
flow we have here
is in in this physical system is between
internal states and external states and
they
so internal states act on external
states via
active states and external states act on
internal states via
sensory states okay now what happens if
we simulate
this such a system in a digital computer
now this is a caricature
or a simple visual model of
a computer with a von neumann
architecture and one of the

one of the characteristic features of
such a computing device is that
it has a distinct memory unit which is
separate from the
cpu of the central processing unit so if
you simulate
a system with this causal flow in such a
system
you'd have to store the values of these
variables
of external internal and blanket states
within the memory unit and in order to
compute
updates of these variables you have to
engage the
cpu so the the main causal flow
would be between the memory unit and the
cpu of course there are also
variants of this architecture in which
the cpu
contains a memory unit but then the main
causal flow would
be within the cpu and
also between the memory unit within the
cpu
and the rest of the cpu so
it's a different kind of causal flow in
this physical system
although the computations
performed by this by these systems
can be the same in the sense that um
in internal states can be interpreted as
conditional probabilities of external
states given blanket states
okay um so this is we suggest one one of
the differences between
simulating and instantiating
um consciousness that there's a

constraint on the
causal interaction or causal flow
between the system
or the system's internal states and its
environment
and of course this may be different in a
computer with a neuromorphic
architecture
okay then in the final section
we relate this
these ideas to our previous work on
minimal
unifying models of consciousness and so
just explain this very briefly
on in general what can computational
models bring to
bring to or how can they facilitate an
understanding of consciousness
so computational models may serve as a
bridge between
phenomenal properties of experience and
properties of
neural activity they may operationalize
theories of consciousness they may
explain
theological functions that is cognitive
capacities
associated with consciousness and
um so these are things that for instance
active influence models
could provide and the
free energy principle together with such
models can then
serve as a unifying framework and how
can it be
unifying so here's just one possibility
um i'll start with what jonathan
burge calls the facilitation hypothesis

in a recent paper
so they're here right phenomenally
conscious perception of the stimulus
facilitates
relative to unconscious perception a
cluster of
cognitive abilities in relation to that
stimulus so this is just a
hypothesis which can be empirically
investigated you can for instance
look at different non-human animals
and see whether
non-human animals that have one of these
cognitive abilities in that cluster also
have the other cognitive abilities
and whether animals that do not have
one of these abilities also lack the
other cognitive abilities
and so this would then corroborate the
hypothesis that
conscious that is that there's some
mechanism that facilitates
or enables all of these or an entire
cluster of cognitive abilities and this
could be
what consciousness does um
in addition to that one could also find
theoretic theoretical supportive
evidence
if one builds minimal models of
these cognitive abilities and one could
show
that a minimal model of one cognitive
ability in that cluster is also a
minimum model of other abilities or
whether
or if one could show that there's a
common

computational mechanism that
accounts for all of these cognitive
abilities and
a hypothetical example just to
illustrate would be
active inference models um if
active inference models of
such a cluster of abilities or required
deep temporal processing or maybe deep
counterfactual processing or something
like that
so if one would have to posit a certain
generative model with a certain
architecture in order to account for
these
different abilities and that that could
be something like a common computational
mechanism
which then could
further corroborate the hypothesis that
there's something that
that facilitates these that jointly
facilitates these cognitive abilities
together
um okay and this this could then lead to
something like a minimal unifying model
i introduced this
term in a paper and neuroscience of
consciousness
a minimal unifying model specifies only
necessary conditions for consciousness
it involves determinable descriptions
that can be made more specific
and unify existing accounts of
consciousness
and i would now add that it can also
unify cognitive abilities associated
with

consciousness not just accounts of
consciousness
not just theories of consciousness okay
but we can
talk about this in the discussion and
the a general picture would be that
there are these different aspects of
the signs of consciousness there are
theories of consciousness
potential teleological functions of
consciousness
that is cognitive capacities associated
with consciousness
and there are mathematical functions
that are computed by conscious systems
there's the notion of complexity here
and then neural
corners of consciousness and perhaps
minimal unifying models could um be
useful here as i've suggested
and the free energy principle can
provide a
way of relating these different
notions and different branches of
research
together for instance active inference
models
can be can be
used to develop operationalized theories
of consciousness
or i mean there's been this
recent active inference model of the of
global
of the global neuronal workspace theory
of consciousness
and um active inference can be used to
model cognitive capacities
and saron and

[Music]
relating these different
notions and these different kinds of
work together would then
could then also lead an insight into why
different
neuronal structures are associated with
consciousness
and what play compact what role
complexity actually plays
in consciousness and
many other questions
and problems okay so this um is the end
of my short presentation
uh thank you for listening and i'm very
much looking forward to the
to the discussion
thank you thank you for that that's
really awesome
so you could great
well what a awesome way to begin
this conversation with a presentation
and
um also really appreciated that you're
revising the paper and kind of giving us
this glimpse of not just a research
field that's
in evolution but the paper itself
is um undergoing changes so for
the rest of this um active stream 16
we'll kind of play it by the usual rules
if anybody wants to just
raise their hand we'll get to everybody
who
does um also we do have the slides and
blue who's um here and i had done 16.0
so we had thought through a couple of
topics so

um at this point i guess we'll open it
up to
people who want to raise their hand and
can also just say hello
on their first time speaking and then um
in the absence of hand raising we'll
just continue through these slides so
thanks again wancha and let's go to
stephen first
and then anyone else who raises their
hand
yeah just like to say thanks very
interesting to hear the developments on
the paper
um i was just going to ask about the um
well i suppose there's a lot of things
that come up but i'm stephen by the way
just say i'm based in toronto
but i'm working on a practice-based phd
through the uk
and one thing that came up on your
slides there was interesting
when you were looking at the potential
for the
there to be some sort of memory that
relates to those states
um and i was curious whether
that memory itself
might have to be connected to sensory
states
or whether that might be something that
which is standalone and if so
whereabouts you might think that might
be
in the brain like is it a hippocampus
type question
grid cell type question or something
like that so thank you

thank you and that's a very interesting
question and
i think it could be useful to show my
slides again
let's do that
okay so you're referring to this picture
right
um and um yeah so
here the idea would be that um what
in a simulating in a system that
simulates
a system that is in a action perception
loop
um there would be some
some internal states that encode the
internal states of of this system
that is being simulated so this digital
computer would have some
physical states that encode the internal
states of the simulated system
and um so
coming to the first part of your
question that there would have to be
some
interaction with sensory input but what
a sensory input so this
in principle one could say well any of
the
other physical states of this digital
computer
could serve as a sensory input
to these internal states right
and the question is what
the causal relations between these
physical states that encode
internal and sensory states of the
simulated systems
what these causal relations be

similar to the causal relations we see
here
and i would suggest no
um and then there's
another aspect you mentioned in your
question
which is whether there could be some
memory unit in the brain and
um so perhaps you can
elaborate a little on this you were
you're thinking about
simulations that are going on in the
brain
or yeah i was kind of thinking
um or curious if you could have maybe
two parallel kind of
you've got the real time real
sensory input being like experienced
and then there's like an update to
this kind of memory but that memory
doesn't
you know that memory rather than it
being a representation
that memory it could be different and
with that memory i'm curious where that
memory itself
then connects back to the actual sensory
organs
you know so it's not like it's in some
computed form
um but somehow the the sense of smell
the sense of
the multi-sensory integration is still
happening at some level you know
but maybe in a different way which then
helps
in dreaming and stuff like that
yeah so this is really interesting um

i mean you i think it
could be useful to distinguish different
types of simulation that could
occur in the brain one would
maybe be involved in um
imagining certain kind of factual
situations
and another would be involved in
dreaming
and you're suggesting that there must be
some
memories somewhere and
which maybe how
which would yeah
there have to be some states that
that are um that give rise to these
conscious experiences but they're not
directly connected to
sensory input is that what you're
suggesting yeah they're not maybe not
directly
connect they're connected to the sensory
organs
and so so it's not that it's like a
separate
it's like you in a way i'm saying it it
needs to keep
some connection to the sensory
what we call inputs i suppose but in a
different way so it's not like
it's it's completely offline if that
makes sense
but it might be online in a different
way
which would then still allow you to have
a dynamical system in both cases
and applause will root for how they came
about um

so yeah so i'm curious about that so
basically
um you you know whatever these
priors are this they're still somehow
um engaged
in the sensory organs
in in being alive and conscious
thanks steven um wanja go for a response
there and then we'll continue with adam
and marco and then anyone else
so the one thing that um comes to mind
here is
that it may be useful to
talk about nested markov blankets um so
you'd have the the brain
with uh sensory states and active states
and of the brain from in states of the
brain would
can be regarded as internal states but
within the brain there would then be
another markov blanket and internal
states within
these internal states which
um would then
be directly associated with
simulations or with dreaming or
other kinds of experience and
so the that there would be an indirect
connection to sensory organs
but the
as it were the sensory states that
immediately
affect these internal states that are
associated with
from say dreaming would
um would be within the brain
are not directly connected to sensory
organs

cool thanks a lot um adam
at denmark
uh can you hear me yep uh
hi everyone uh hi wanja thanks for that
that was
this is really fascinating work and it's
uh disturbing me
and that i've recently attempted to
present a mineral unify model uh
using the free energy principle as a
overarching framework
and i'm wondering whether i need to
backpedal on some of the visit the
claims i made
so uh i tried to bring together um
integrated information theory and global
neural workspace theory
and but in doing this one of the things
i pushed back against an iit
is uh sorry i'm not tried my little
breathless
um one claim i was trying to push back
on is that
uh you you needed a neuromorphic
architecture
um to have consciousness and that you
couldn't do it
um unavoidable i think this issue is
probably
largely moot um moot
do the just what probably would take to
get
enough compute we probably need
neuromorphics just for energy efficiency
but um i'm wondering
um with respect to that that slide
and in terms of causation whether
like by the time we commit to a core

screening
that involves those entities that
actually um we might need to coarse
grain over
the workings of the vinom architecture
and actually go into the virtual machine
and that the the causation is
in terms of like a like in perlian
causation we have to like
commit like commit to an ontology and
then once we have our directed graphical
model then we're doing the operations
and
that's like our causation and so to i'm
wondering if there might be like a
mixing
in that figure and that if we're like i
don't know if that makes sense
um but and i guess with relation to um
nested markov blankets
i i and and steven's question um also
i'm wondering if this might end up
taking um
biological consciousness and potentially
putting it in the same kind of
epiphenomenal boat
in terms of degrees of how much
directness
allows us to um
actually treat uh sensory motor coupling
as actually being
uh real and causal so i'm wondering if
we
do that we end up like being in the same
boat of
uh not ascribing consciousness to
um biological systems either even though
they're

using massively parallel operations and
they are
in their very dynamics having this dual
aspect character
i do not know cool launcher
go for it and then uh marca thank you so
much
um by the way um adam thanks for
attending this session
so um and you had some immediately i had
some
comments on twitter when i posted a link
to the preprint
and i didn't have time to respond to
these comments so i'm really grateful to
you for
attending this so we can um actually
just discuss the points that you
raise and yeah so
how real our virtual machines and
i think there's two issues that have to
be separated
and so this is one thing i'm sure about
and then there's one thing that i'm less
confident about so the thing i'm
sure about is if you
have a course grain description of a
system
and abstracting away from certain
details
that doesn't mean that you end up
describing things that are not
as real as the things that are happening
at a
more detailed level um so if you're
describing a brain in terms
of the what's going on in individual
neurons and describe that in

a lot of detail that's more specific
description you have than if you're just
um considering the average
activity of neuronal populations
but that doesn't mean that what's going
on at a
population level is less real or that
these average this is
this average activity cannot
um have causal powers or that
relations that you would describe that
way that you
could describe as causal relations at
this more coarse-grained level
that these relations are not really
causal relations or something like that
so i think we agree on this part
and then the question is well if
at such a course grade level if i
understand you correctly at this
at such a more coarse grained level if
this instantiates a virtual machine
doesn't this mean that the virtual
machine is
um as real as the physical
machine that instantiates the machine
and
i actually i don't know but there are
some things
that follow if you accept that virtual
machines can be conscious that are
really really counterintuitive and
disturbing
so um i mean if you
if you if you what if you've watched a
series black mirror
there are these episodes in which
a um person is uploaded to a computer

and then there's a virtual version of
that person in a virtual environment
and that virtual person is being
tortured for
i don't know for weeks or even months
and the assumption is that this
virtual entity is conscious
and this happens in
over a course of i don't know minutes in
the external real world
so um if if a virtual machine can be
conscious then
conscious experience is that we have
hearing during
over the course of several weeks or
months could
take place within minutes or maybe even
within seconds and that's really strange
but you could also run these
a computational processes backwards
and then you'd have backwards
conscious experiences or you could stop
them
at any moment and then
continue them and or you could have
physically highly distributed systems
that um are connected by
some some device
which then enable this enables
this collection of physically scattered
systems
to compute some to perform some
computations
and if this can then give rise to
consciousness
you'd have a very strange conscious
entity
and other things like that and then

there are these thought experiments such
as
uh net blocks blockhead
experiment in which the chinese nation
instantiates consciousness um or at
least the
functional states that um
may be associated with consciousness and
yeah so there are these really
counter-intuitive
consequences and
this is the main motivation for me for
trying to
make a distinction between simulating
consciousness and
instantiating consciousness and
maybe there are some virtual machines
that
are conscious but
i my intuition is that it would have to
fulfill very specific conditions
and one of these conditions may be that
the interaction it has with its
physical environment must be the same
type of interaction that the
simulated system has with the simulated
environment and that's what i try to
illustrate here but of course you could
you could argue well this is just
motivated reasoning because you
don't want to accept these
counterintuitive consequences
and why what would you make
make this accept why should one accept
this
requirement
thanks it's an interesting style the
argument from consequences with

consciousness well that can't be correct
because then this would be conscious and
we know that's not true it's like wait
do we know that so marco and then blue
oh yeah another question another
comment about a nested mark of blankets
and
um so you suggested that
maybe this requirement is too strong
if we maybe it's fine if we apply that
to
um to digital computers
but um if we apply that to
actual to biological organisms
that we would regard as conscious
and the result would be that they are
not really conscious at all so
it's too strong i guess that's what
you're suggesting or what you are
um the question that you're raising
so um yeah
i guess one would have to flash this
out in a bit more detail but one
way would be to say look the free energy
principle makes some
idealizing assumptions and the very
notion of a markov blanket as used
in free energy principle is as um
i think many know by now uh slightly
different from the
notion used by or proposed by judea
pearl
um because pearl's notion of a mark of
blanket it's just a
feature a pulse markov blanket is just a
feature of
a model of certain
causal graphs and

the markov blanket used in
the free energy principle is an
ontological
notion or metaphysical notion because
it's
meant to be actually part of a physical
system
and um one good question
that it really
exists and um
yeah so or that
optional biological organisms that are
conscious
that they act actually have these
or that they fulfill these idealizing
assumptions
made by the finishing principle and then
for this reason
if we derive a constraint on what it
means to be a conscious system
as opposed to simulating a conscious
system from the free energy principle
actually requiring that a system
has these or fulfills these
idealizing assumptions which
may be features of a model of a system
but
are not actually satisfied are not
actually instantiated
by the system itself um
yeah i think there's an ongoing
discussion or there will be an ongoing
discussion about
the notion of a markup blanket and
whether it's justified to treating it as
a
ontological notion as a
feature of an actual physical system

um and we are committed to this strong
reading and in the paper so
we're using the notion of a markov
blanket in the
ontological sense not just in the
epistemic sense
so this may be a problem
cool thank you for that edition so marco
then blue
thanks um thanks for the uh presentation
that
rates that paper i've been significantly
excited
by reading it especially a great uh use
of fep
uh and buyers because it particularly
it's a lot of intuitions i had
um and it's also very nice in the spirit
of pluralism right this approach of
really embracing the unifying quality or
power of the fdp and focusing on
unification rather than
another universal or overly saturated um
claim proposition um and it's going to
be brought up some scattered thoughts
but i'm trying to
also touch on what stephen said i'm not
sure if i understood what it meant
um but i'm reminded of the amortization
so
so sam gershman has some nice papers on
this uh he has a nice paper called
uh remembrances of influences past
so the parameters or the the components
the factors of the strategy
used in influence could also be
remembered and so maybe that's
what i thought you meant when you said

something related to or
associated with sensory states but not
actually synthesis itself
maybe in the physical substrate that
enacts these sensory inferences
and there might be partial many parts or
parameters or factors
um that can be remembered and that these
are the ones can be
these are the ones that also are
reactivated during dreaming despite the
absence of sensory inputs
um but but anyways if it's not what you
meant then
let's talk about that later um
uh about about your conversation with
adam i um
i have a bit of a i'm not sure if you
meant virtual machines as
popular as uh in actual computers and
software
for me it's very important to to
distinguish a mere embeddedness
uh such as the normal virtual machines
and an embeddedness where the
where the relation between the embedded
subsystem and the larger system
is one that's related by active
systematic
mutualistic influence right that these
are mutually contextualizing and there
is a kind of targeted relation to it
which is not the case
with normal uh virtual machines
especially in the norman architectures
because these are more sterile channels
as i would like to say
um so as for my own points i've been

reading it so so
um i very much uh appreciate that you
elaborating emphasizing the notion of
dynamical complexity as opposed to
complexity and the and very elegantly
avoiding the many conceptual traps
that tends to be lurking in these woods
um
but but but i have a suggestion or
question so
so you
suggest that the notion of accuracy kind
of falls away in the determination of
free energy
uh variation free energy um for honest
awareness
but i actually would argue that that in
fact it accuracy is still relevant but
it's a reflexive and of accuracy
so because of the embedding this or the
mark of like it's immortal magnets or
the embeddedness of these ecosystems
um you could say that that to some
subsystems their
concerns of accuracy are relatively the
other sources which are
eternal system and so you see this in
dreaming right so so the
so the different subsystems of the brain
need to update themselves and each
in relation to each other and so for
example um i think tononi has some work
i'm not sure if it's published yet
but that the activation of uh during
dreaming
or the epicenters of the activity
radiating outwards
seem to correlate with the sites of

activation during
the waking experience where something
new is learned and so this
and and under the sap it becomes
extremely clear well it suggests
an intuitive nature for why because if
there's a local kind of
improvement or local updating occurred
in that model
but all the substances in the ecosystem
are mutually contingent or
contextualized by each other
then it would be useful to kind of
extend the fruits outward or share the
fruits of that experience with other
systems
and so you get you also then lose the
the necessity of a global coherence of
the ecosystem
because there is no uh unified external
world right all the contexts become
local
it's simply a constellation of local
continents
which in my opinion is an elegant way to
look at it because it seems parallel to
the phenomenology
of dreaming which um you know argues for
your paper's motivation
because this tentative computational
explanation
um has in uh immediately a nice pathway
to the phenomenological uh aspects to be
explained
um so yeah so i would just really like
to uh
hear your thoughts about but whether out
should really be dropped or whether we

should see
of still being a part of the variation
energy in the lower levels
so more specifically it's as if the
complexity at one level
is partially constituted by the accuracy
improvements of the lower levels right
and so
especially because you you you mentioned
embedded this but i don't see it as
well integrated in the paper but i think
it would
much more bolster your claims especially
because of its relations with other
material and consciousness research
especially before ecamp and their whole
emphasis on
embeddedness um and and this also
relates a bit
well i would say that that in another
way you can see this as a contextual
reduction
can be seen as uh configurational
accuracy
so if we assume that that's the act of
choosing the configuration
is itself a question of policy selection
or the action
of configuring yourself in relation to
the context then complexity can be seen
as a configurative accuracy right
um and so now you get these weird things
so decomposition
of uh complexity reduction in terms of
intrinsic context reduction
so kind of the bayesian model reduction
you see scene the active pure active
influence of curiosity paper and an

extrinsic model reduction
a modern reduction in relation to the um
contextual
systems in which it stands relationship
anyways i'll stop here i think
i'm rambling on too much so um yeah we'd
love to hear your thoughts
so um thank you so much for your
comments and and questions
on yeah so this point about
embedded systems and nested markov
blankets that's
definitely something that i have to
think about a bit more which will be
relevant relevant so when you talked
about complexity
and how within a an
embedded system within the brain it
could still be
um accuracy
maximization despite
a lack of sensory input
via other sensory organs
i actually wasn't reminded of a
discussion we had
with a reviewer
but not a reviewer of this
paper but of the
markovian monism paper we published last
year so
one of the reviewers wrought
brought this actually this this question
up what happens
in a an isolated system
and they mentioned that according to iit
it i i think it could still
be conscious something like that it
really reminded me of

what you said or the other way around
what you said reminded me of this
discussion we had and the my
intuition i had or the the initial
reaction
i had was that okay well in such systems
we have
maybe no sensory input from
the from the external environment
but there's some environment within
the system within the organism that can
provide some simulated sensory signals
and i think this is what you were
referring to
and i don't really remember what
happened
i think carl had a slightly different
opinion
regarding this and suggested
that the most
the more elegant way of treating
this case would be to regard
um the system as a system that
just minimizes complexity and it ignores
accuracy but
now that it comes up again i don't think
that's the end of the story
and it may be
relevant and interesting to make this a
bit more
complicated by actually considering
nested systems and simulated sensory
signals and so on and i don't think that
this is something we
can build into the revisions because
the paper is already getting quite long
but it's something that we should focus
on

in in future work and um
i would be grateful if you could um
point me to some of the literature that
you mentioned
um send
and then there's another thing you
mentioned a distinction between
two types of virtual machines or two
sensors in which
a virtual machine can be under
things that can be associated with that
term
and um i
i don't think i am um aware of this
distinction and
i'm not sure i already understood it so
perhaps you can
repeat what you said um so that one
is nearly embedded and the other
has some some additional features
i think that could be really uh
interesting and and relevant
should do an hour later yeah go go for
it the keep in thread and then
we'll go to blue and then to dave and
adam
and another question from chat i try to
keep it brief so
so so actually one of the reasons why
i'm so motivated about this is actually
because when i did my master thesis with
carl
i had the same argument complexity so
he's also like no no it's just
complexity reduction
but but the thing is as you know it's
about redundant parameters but i don't
think that's going to be the issue in

the story that i told you
it's not about redundant parameters it's
parameters that are mr tuned
because of their contextual
contingencies having been updated
but they themselves not because in the
real life awakening experience it wasn't
they weren't interacting but in a more
extended context it's kind of
a weird kind of counterfactual
marginalization but but it's like
expanding the scope
in which the newly accrued experience
might be rather easy that's how i see it
and so it doesn't seem to me that it's
valid to say it's merely about
removing redundant parameters um it is
improving the configurat configurative
preparation
for anticipation in the future given the
limited scope of the experience that was
newly acquired
as for the virtual machines one
important uh difference is for example
the
the the mutualistic relation between for
example
merely the transistors and the virtual
machine they're in
um it it's it's um kind of
flat or low dimension there's no
multi-skill multi-level thing there's no
adaptivity all the way as in the driving
uh the driving force behind the
generation of
the virtual machine virtual machine
itself is is kind of uh
is is is simply sent out

it is not in the path itself it's not uh
contingent upon the path integration
so so there is this non
is it's not an active integration of
their relation okay never mind
i'll try to think about how to phrase it
but my intuition says there's a very
important difference between virtual
machines
classical form and the kind of
virtuality that you that is relevant for
minds but especially consciousness
thanks marco for these awesome points
and that's the fun of the synchronous
moments that we share is to surface all
these new connections and ideas
and then we return to the work and the
sharing on
you know platforms with each other so
thanks again for that ronda if you
wanted to add anything
otherwise we can go to blue and date
okay awesome so we'll go blue dave and
then we'll have heard from everybody
once
then the chat and then return to adam
and everybody else
okay so i want to go back to um
your discussion with adam also uh and it
incorporates um virtual machines really
so you were discussing simulating
consciousness versus instantiating
consciousness
and the necessary conditions for both
and in describing like the black mirror
episode which i've seen and is awesome
um you know you really implied that
there's this necessary

like arrow of time going forward
for maybe instantiating consciousness
that's not present
in simulating consciousness and so is
that
like one of the required conditions like
does the arrow of time
play a role like this bi-directional
like time
is like not that that can't be conscious
and
maybe it would also help um and i'm
sorry if i missed this in your
presentation
because i was a few minutes late but um
i i don't know how exactly you define
consciousness
and i mean i did the dot zero with
daniel and
it's really kind of difficult to um you
know measure
or ascribe scientific properties to
something that's like
not clearly defined and i know that
there's many definitions but i just
was wondering wanja what you personally
take consciousness to be or how you
understand it
okay thank you very much for these
questions um so i start with the second
question because that's
um in a way it's easier to answer that
question
not because i can define what
consciousness is but because i can
just tell you that i don't know what
consciousness is
um so i mean there are

various ways of operationalizing
consciousness
and ways that are used in
research on the neural habits of
consciousness in particular
and many of these
measures of consciousness involve some
kind of report
rubber or behavioral report and uh
for instance the question is than just
does the can can the subject report
reliably report the presence of a
certain visual
stimulus or not
and that would just be an example and so
i don't have a strong opinion about ways
of
what's the best best way to measure
consciousness
um but
so what we are kind of
uh building on here is just the existing
work on new
colors of consciousness which use
various ways of operationalizing
consciousness various ways of measuring
consciousness whatever it is
um but of course this is a bit
unsatisfying
and we would like to know what is
consciousness we perhaps
you can tell us how to measure it in
different ways but
what is it that is being measured and
here i actually think that a
computational explanation of
consciousness could
provide an insight into the very concept

of the notion of consciousness
in that it if if there's a
common computational mechanism
associated with different
cognitive capacities that are
facilitated by
what we call consciousness and
this could provide part of
the puzzle that gives us a definition of
consciousness
so just as an example um think of
the global workspace theory according to
which
consciousness is basically global
availability
well this you could use that as a
definition of consciousness
consciousness is
of global availability to
different cognitive subsystems
so information is process consciously if
it's not just
available to um some
individual subsystems but to
all or most of the cognitive subsystems
within the system and um yeah this is of
course a bit
unsatisfying because it seems
that it doesn't capture everything that
we associate with
consciousness so consciousness has many
interesting features um so if we
try to describe the phenomenology of
consciousness
then we can say that it's typically
structured
we experience spatial relations
between things so you're not just

experiencing
say a desk and a
computer screen and whatever
else you're you're consciously
perceiving right now
you're not just hearing something but
you're experiencing these things in
relation to each other
and there are spatial relations that
you're experiencing
they're pothole relations and there are
also temporal relations of course you're
experiencing not just isolated events
but successions of events
and it seems that what is happening
right now
is not just a an instant but it's
somehow extended or a um
yeah a specious present
as william james called it
and and there are further
general structural features of
consciousness
and what a definition of consciousness
should provide
is a means of making sense of these
different properties
so for instance
if consciousness is global availability
of information
in a cognitive system why does this mean
that
why does consciousness have
these phenomenological features
that characterized characterize most or
maybe all
types of conscious experience so that's
something that a

definition of consciousness should
provide
in addition to just saying
what's happening in the brain and
computational models i think could
provide
a means of
breaching levels of descriptions
let's say
descriptions of neural activity and
neural mechanisms
and then phenomenological descriptions
that um describe
features of consciousness as they appear
from the first person perspective
and integrated information theory is of
course an
example of a theory that tries to
achieve just that starting with
phenomenological
axioms that characterize consciousness
or
typical forms of conscious experience
and then trying to
operationalize these axioms
in a formal way such structure they can
be
applied to your
neural structures on your activity
or other kind of system and
in principle i i think this this would
be how one should
try to define consciousness
um finding computational mechanisms
underpinning cognitive capacities that
are associated with consciousness
and trying to relate
models of neural activity or maybe

computational models of cognitive
capacities
two characteristic phenomenological
features of conscious experience
and then yeah so that this is
what what i think how one should
approach the question
what is consciousness can i ask a
follow-up really quick
yes uh so i just wanted to um
so you're just you're describing really
this global availability
and like able it's it sounds a lot like
um
david krakauer described individuality
like in 2020
with with like this bi-directional
information flow
in in an individual right but there was
no
like conscious um so you have to have
that downward causation which is which
sounds a lot like the global
availability that you're describing and
so i just was wondering i don't know if
you've read the paper or not
but um it also like i got that i got
that out of what you were responding but
i also
tended to think that um you're maybe
only talking about human consciousness
but then when you went into iit maybe
not
so so do you actually draw a line like
what
what kind of the system can be conscious
does it have to be biological
does it have to be like higher order

vertebrates or
um can do you think that this is
especially in terms of computationally
like defining consciousness which i also
think is a great place to start
um but does does this mean that maybe
all systems up and down the um you know
evolutionary ladder can maybe
have some degree of consciousness or
what do you think what are your thoughts
on that
thanks for the question and so
where to start maybe first um you
mentioned top-down causation
i wouldn't assume that consciousness
involves
top-down causation so i i don't see a
reason for assuming
that and that's not what i meant by
global availability um
so um i i'm not familiar with
this work by david krakow that we
mentioned
but um it seems to me that it may not be
um may not be directly connected
what i was trying to
um and then um
the question where where to draw the
line between conscious and unconscious
questions
uh systems that's i think a really
interesting
question and a further i mean i'm
in the beginning of the talk in the
beginning of the presentation i
mentioned
one motivation for um
going into research on computational

correlates of consciousness and that was
trying to have some um
correlates of consciousness or criteria
for consciousness
maybe necessary conditions for
consciousness
that can be applied not just to
human beings or to human beings in
ordinary states on or human beings with
a
normal brain as it were but also to
unusual cases
and i in the presentation i only
mentioned
the possibility of islands of awareness
isolated systems but of course
unusual cases also include non-human
animals that are
physiologically different from
human beings invertebrates
artificial systems as well and
this is largely i would say an empirical
question where where to draw this line
and there's recent work for instance by
um eva blanca and simona ginsberg
who um have a
an evolutionary hypothesis about
consciousness or hypothesis about the
evolution of consciousness and they
suggest that a particular type of
learning
that is a learning capacity can
maybe the evolutionary transition marker
of consciousness or
creatures that during
a species that um developed this
capacity were were then also conscious
and the capacity learning capacity

they call unlimited associative
learning which
involves particular
types of learning such as cross model
learning and other types of learning and
so their suggestion would be to
look if there are what systems
are capable of this or have this
learning capacity
and what what other
um what other capacities
cognitive capacities they have and
um so this is
basically an empirical question trying
to determine
which creatures are conscious and which
are not and
we of course need to have some criteria
that can be applied to determine whether
the creature is conscious or not and
if we don't have a definition of
consciousness or a
theory of consciousness that can be
applied to non-human animals and it's
really difficult to tackle this question
but focusing on technological functions
of consciousness and
cognitive capacities of consciousness is
one
i think particularly promising strategy
and i was
also mentioned jonathan birch who um
has proposed this that there may be
clusters of cognitive capacities
associated with consciousness such as
trace conditioning or cross-model
learning
other capacities um

yeah so i i don't know where where to
draw the line i think it
is an empirical question and um
these more empirical projects
trying to investigate
cognitive capacities of certain types of
non-human animals
can i think be complemented by
theoretical research
using computational models because
that's one of the advantages of
computational models that it can be
applied to different types of systems
not just these systems that have a
central
um nervous system such as ourselves but
um also systems that have a
different physiology or in principle
also
artificial systems i wouldn't assume
i would make the assumption that
artificial systems um cannot be
conscious but i think it will be
really relevant to determine
what constraints what conditions have to
be fulfilled by
a art by an artificial system in order
to be conscious
and which systems which types of system
will never be conscious
and if for instance this black mirror
scenario
is a a
physical possibility if we could
actually build such
devices that simulate conscious beings
and thereby create
real suffering for instance that would

be horrible
and it would also mean that we could
just
make hundreds of copies of these virtual
systems
and instantiate the same type of
um maybe horrible conscious experience
in multiple times that would be
a really um a disgusting scenario and
so i think it would be
it's extremely relevant to
investigate the difference between
simulating a system
and instantiating consciousness to avoid
creating unnecessary suffering or
disturbing conscious experiences and um
you had a very interesting question
about the
arrow of time going forward and
yeah this is something that i guess i
have to think about
in a bit more detail so what what
happens to the
causal flow or the action perception
loop if you
reverse the temporal direction of the
simulated system and um
so intuitively i would say that it's
important whether the arrow of time
goes forwards or backwards that you
wouldn't
have conscious experience if you
rewind the experience of
a simulated system
but i'm not sure about that it's a
really interesting question
thank you thanks for the response so
we're going to do

dave then a question from the chat then
adam and marca
it sounds like getting a handle on
consciousness maybe
suffering because it's been deflated too
much
if you assume that something you're
examining is a
predicate it gets really easy to turn it
into not just
a predicate but a thing and things
are pretty hard to work with mostly
they seem too easy um
the
first first over deflation
is assuming that conscious acts
or conscious activities are
persistent long-term things a predicate
of an organism
the organism is present now if you if
you subscript that and say
is present of certain things now and
then
under some conditions well then you can
ask what conditions enable consciousness
what's achieved what's the evolutionary
advantage of being able to be conscious
when you need to
and people as disparate as gerald
edelman
talking about conscious flashes in the
brain the
the dynamic core crystallizing now and
then
and then there's conscious activity and
then maybe it's not needed because the
the problem has been resolved earlier
alfred north whitehead talking about

threads
of higher experience within a brain
for instance something that happens now
and then
uh and then the transformational
psychologist gf
actually gives an exercise to force
people to be conscious and he said he
his background is most people are
non-conscious
the great majority of the time however
if i ask you
are you conscious the answer is always
yes because i've made you conscious by
asking you that specific
leading question
now the notion that consciousness is
something that comes into you from the
outside that's been around for a good
while
that's what the physiologists in these
1870s and 80s assumed
so freud since he was busy thinking
about so many things didn't want to
tackle that and he just
assumed oh yeah consciousness comes in
something's going on
the outside of my brain my neocortex
responds to that
and that's where the consciousness comes
from
and if i'm not being stimulated well
i enter into nirvana i don't
dislike and i don't like i'm just
in a great blah state
and the engine is just idling and
waiting for something else to make it do
something

so oh and the other way conscious
activity is getting deflated too much
is as i said being treated as a
predicate
well it seems that
if you expand it to a relation
it might be a lot easier to start
dissecting
i'd say off the top of my head a
relation among 10 terms
self a is conscious
with self b of object
x in that conscious relation between
a and b of x given
the presuppositions of a b and
the emergent relation
and the three
sets of dissatisfactions or goals
uh whether it's uh an unpleasant
stimulus wanting to get away from
something or
getting towards something of a and of b
and of that emergent relation between
the two so that's ten
a ten place predicate right there and
when i'm thinking hard i don't really
see any of those ten
dropping out because when i'm in dial
out dialogue with myself i'm at least
that complex and is so so is my dialogue
with myself
how about you
thanks dave for the question uh laundry
and then i'll ask something from chat
thank you um yeah so you made some
really interesting remarks and
in fact i i think it's important to
see consciousness not just as as a

property that an organism
has um static
property but as something dynamic that
can be transiently lost and so on
and um so your
suggestion to suggestion to treat
consciousness as a
phase relation um is pretty interesting
and
i think this speaks to the the idea
of investigating nested markov blankets
and interactions
within between subsystems within the
brain or within
conscious organisms and
i think you're completely right that a
lot of the
relations that can hold between
um a conscious being and other conscious
beings or
conscious beings and parts of the
environment
must in some way be mirrored
in mirrored by relations between
parts of the system of
subsystems of the system
and yeah these are
really complex issues that we try to
um try to
try to ignore in the paper but which i
think we should
get into at some point especially the
idea that they're
on nested systems nested
mark of blankets within the brain
so i think this is really important
thanks for the response so i'm gonna ask
a question from cambridge

breaths in the chat and um just for this
last you know
25 or a little bit longer i have the
slide up for 16.2
so we don't need to worry about
answering consciousness in the next 20
minutes let's actually
build the energy for continuing our
discussion as we do
in between the weeks so cambridge
breath's question was and this is
related to this discussion
does every memory or thought episode um
have a markov blanket of its own that
can be mathematically characterized and
measured
in other words can we consider every
thought episode
every minimal phenomenal experience as
an independent island of awareness on
its own
so wanja i'd love for you to give a
thought on that how do we
operationalize the markov blankets are
we a singular blanket or is it just this
constellation
of little transient bubbles and how does
it relate to this islands of awareness
idea that you mentioned in the paper
thank you um yeah so i'm not sure how to
um how to formalize a memory or
thought episode but i'm sure it would
involve some
mark of blanket but that doesn't mean
that there's an
isolated system or island of awareness
within
a conscious system because there

would be some interaction with
the internal environment as it were
um another question that
comes to my mind now is that if if you
um imagine something i have a thought
episode
there will be some part of
and if we assume that some part of your
brain
realizes this broad episode
what would there be a conscious system
within a conscious system
because of it if it has the right
engages in the right cause of
interactions with its environment
and i think this
should be added as a or could be built
into
a further constraint of what what it
means to
simulate as opposed to instantiate
consciousness um i'm not sure how
exactly but
so no i wouldn't say there's an island
of awareness
when you have a thought episode or
remember some events
um but there are some really i think
relevant points associated with this
yeah
thanks and it's just funny how we talk
about instantiation and for instance
and we talk about instances of virtual
machines in the linux
sense so it's actually a nice
computational um
language so i have adam and then
marco than anyone else who wants to um

raise their hand so adam or yep go ahead
uh hi can you hear me yep as usual
stop asking then so um
i you mentioned like with the reviewers
um uh so like
you have these like lists of like places
where like your intuition is balking
and i i share a lot of those like um
could a nation
uh be conscious could like um so a bunch
of tin cans and ropes strung together
and there's a sense in which like just
like church tearing thesis i guess like
there should be a certain universality
of
computational realizability and maybe we
can do like
um in in principle tin cans and
string but it just might not be
practically realizable because
there aren't enough you know there's
enough aluminum in the multiverse or
sorry in the universe
but i actually liked your proposal um
that
like you kind of dismissed it as like
maybe being like a little bit special
pleading
but i didn't find that to be the case at
all so like if you actually
created a rich embodiment in embedding
within a virtual world and then if
there's some sort of like homomorphism
between this and
i guess i actually i don't even know if
it needs to actually couple
with the physical world
like the importance would be that you

you're having this sort of
um functional closure with respect to
active and perceptual states
in this sort of self-evidencing and
self-making that you're having
within this within the system you're
doing inference on this
interaction and
within the context of which um
within the context of the virtual
machine virtual world which
you're running this um
it seems like we could say that
consciousness is arising but
i'm not sure um in terms of
functions of it so i've been wondering
whether we can think of consciousness as
a kind of like
world model and we're doing a divergence
minimization with respect to
and it's having some causal significance
of this kind
and where it needs to have certain um
additional necessary
uh properties for it to be a world model
capable of generating experience
of having um basically coherence with
respect to like spaceships
oral and causation and causal kinds of
coherence and
um i'm wondering if that could
potentially
speak to the issue like to bring up
functions of
which systems um are more or less likely
to possess consciousnesses like
we just add in a additional stipulations
as additional necessary conditions and

i'm i'm basically wondering if like
quasi kantian like a priori categories
or some of the things people talk about
in terms of core knowledge
these might just be necessary for any
kind of
world to appear to any being whatsoever
for any kind of coherent sense making
and with the function being you have you
have um
some sort of a data structure
corresponding to
the agent and the situation to the world
and this is acting as a kind of top-down
controller and in ways maybe is a
virtual machine of sorts
if you want to if you have any thoughts
on that otherwise
um yeah thank you adam um
and does the system need a world model
in order to be
conscious or not and
um so
i think if if it needs a world model it
doesn't have to be a
very compact model
um and
[Music]
i mean it would have it must have a
world model in the sense that
its internal states
must encode probability distributions
over external states given blanket
states
so but but this is not the world model
you were
thinking of right um well
you were i'm wondering if we need a

little more to actually have like
composition and i guess like the kind of
accents
like to have like things related to
things
with particular like distinct properties
um where those properties are situated
for the various things that are in
relationships
and whether this is basically serving
the function of uh allowing you to surf
uncertainty
basically like allowing you to both uh
be informed by and reform these action
perception cycles
and maybe you need additional like
structure built in for this to actually
be like
a world represented by the internal
states
not sure though i i think perhaps um
you don't have to have a model that
contains objects like
um a system that is conscious
you can have a conscious system that
doesn't know what a tree is or what
an apple is but i think you i mean
you hinted at um spatial
temporal and causal relations i think it
must have a sense of spatial and
temporal relations
and causation that would be
candidate i guess so that would be a
world model
which um could be fairly abstract in a
way about
gives the system a notion of time and
space and

causation and this is what
i guess um i mean one thing that
i um completely sidestepped here is the
issue of
benefit has caught minimal phenomenal
experiences that it's
conscious experiences that
are characterized by the absence of
spatial relations
and or even temporal relations in which
there's no subject object structure
structure so um these are
also sometimes called non-egoic states
or
states of pure awareness or
consciousness as such
and are this described in
especially in
literature from eastern traditions
because these are states that can be
experienced during
meditation but also
um in other um during other states
and perhaps you don't need a world model
and these
types of experience and minimal
phenomenal experiences
thanks oh yeah adam go for it do you
think it's possible and those sorts of
non-dual states
that there could be like a very minimal
coherence along those lines like
sometimes like um who's that
gun toku like sometimes it's like
described as like just spaciousness it's
like just
the space of awareness as these stripped
down non-dual states like could there

still be sort of
it's not enough that it's necessarily um
a movie it's not clearly like
identifiable it's like here's a movie
from your point of view and you're
separate from the world and you're an
agent represented but still
there's just enough of a point of view
just enough
um like they're they're in there but
they're not like
reflexively modelable i don't know
um yeah that's a possibility and it's
actually
um a really interesting option so
with respect to a so-called um
non-egoic states of consciousness or
states
which the ego dissolves
there's a possibility that actually
the subject object distinction
disappears so
those are non-dual states of awareness
but not because there's no subject
but because the subject is expanded
and contains
everything there is so i mean
in such states or in the transition to
such states
subjects often describe a
dissolution of the boundary between
themselves and the environment
and one possibility is that what happens
is not that the the word
or that the boundary becomes blurry and
then the self disappears
but actually that i become the world and
so i cannot distinguish myself from the

world anymore
and that's why people say there was no i
so that's at least one possibility when
um
i think sasha i think has argued for
this
point and one could apply the same
idea to space i would say
and that's what would be similar to what
you suggest that if
there's there are no points in space
anymore no place where i am
but if i'm the world then i'm
everywhere so there are no
points in space that i could distinguish
from the point
at which i am i myself am located
because i am everywhere and
have become one with the world
and so there's only space as such
perhaps only the world but no places
within the world no points that could be
differentiated
thanks for the response makes me think
of king of infinite space and
a circle with circumference of
everything and inside of nothing or
however you want to phrase it so um we
got 10 minutes left
marco and then anyone for closing
thoughts and again i have this slide up
where we're writing things down for
next week so marco and then anyone else
who wants to give a closing thought
and there's one last little question to
chat to so go ahead marco
i'll do my best to keep it short um
regarding the last few comments uh

safely say that which is called space
is called by buddha as no space
therefore it's called space
and so your conversation was very much
about the notions of emptiness or the
rate of luminosity
which is a fascinating body of
literature in buddhism
but but i would argue against the notion
that we should
take as valid context the idea that
oh now we contain the whole world
because it's exactly in that
phenomenological stage
where language breaks down and so any
reports of that feminology must be taken
with a heavy great results
right so one alternative is so so the
way i would see it in
in semi fep terms is that it's a state
of
maximum potential reach right so it's
it's not that everything
is contained within that state it's that
everything is reachable from that state
it's like uh it's it's a kind of optimal
readiness
but contrasting that maximum potential
reach is a minimal determinant
which relates to the buddhist ignotion
of no grasping or no clutching right
you're not grasping to any duality any
determination
you're just letting go of any
determination whilst being ready to
determine
whenever it arises or is necessary which
is

related to the co-dependent arising
which again is beautifully
uh congruent with active inferential
tracks um
so uh i've tried to refrain too much
from trying to solve conscience but but
since since something's already been
raised so i think i've
uh pinned down why i talked about the
virtual model in that sense
the virtual machine and so just like um
adam as always i i also agree very much
with
with emphasizing the cybernetics
compared which i found surprising for
you not to have included
from what i saw so far in the paper so
so so um i i think i think one problem
in
the kind of the folks ecology slash
language we're used to
when talking about these things so we
say that the brain is conscious but we
only say that because we don't
have the ability to specify in what way
the brain
generates or relates to consciousness
and so
following david chalmers i would argue
that that it's better to
use something neutral right instead of
saying consciousness
uh we could take seriously the
phonological invariants across people's
reports
and say that this must be some
integrated a reality let's call that a
local space or a locatology

i prefer to call the c space um
and so you could say that that the c
space embedded in the mind's brain
which nicely has an acne within the
marketing
and is that the c space effectively has
primarily a cybernetic
imperative but in the pursuit of that
cybernetic comparative it actually
finds itself doing diff influence
or inferential initiatives simply by the
virtue of the fact that it needs to
bundle its environment
which for the c-space for that's a
sensible conscious space
um would not be the world as such but
but the
world modeling mind's body uh system
and so you get this weird a tripod type
distinction which actually joshua bach's
first one for actually pointing to this
to this idea
so that you have embedded in the
universe in the physical world
a particular system called the body with
a brain
uh with the systematicity that we call
the mind and embedded in turn
we have something called the c space i
importantly
but that itself uh due to its imperative
cybernetics or
good regulation uh learns about that
my body and it's the mutual
contextualization they're in
so upon instantiation or realization of
this kind of
disease space they become mutually

dependent on each other
the mind brains activity or let's say
fitness would translate transitively to
the students of the seizures which in
turn transitively
translates back to the fitness of the
entire mind body and so this is very
important to note the the
the requisites of uh of the problem that
the initial challenge
of creating enough challenge enough
free energy gradients to necessitate
instantiation such as space
and so i don't agree at all with these
unnecessary debates about whether
organelles are goddamn conscious like
what the hell what's going to come in
for them
they have no adaptive imperative there's
no incentive to instantiate some kind of
regulatory space regulatory subsystem
right
there is no coherence instantiated to
even lead to that there's no
initial ignition to scaffold that
complexity
and so i i i yeah anyways that's my
personal annoyance
another notion i would like to share is
the island is the awareness the fact
that the island is embedded in this huge
amorphous sea
is what we should relate to that
seemingly secluded experience
um and and crucially and this is very
intensive but i want to share it because
of your paper so so i
i would argue or propose that the c

space is there where the intrinsic
manifolds of the extrinsic manifold
become equivalent
or at least approximately or or
asymptotically
why because uh it's another way of
phrasing that it's pure reflexivity
those causal flows or spatial temporal
causal
activity that is purely reflexive
therefore would not uh pertain to
directly
that which is outside there in other
words that c space is by definition the
self
referring island and now comes one more
crazy thing because trauma said we need
crazy ideas so here's a crazy idea
um so the point is that it's possible
perhaps
that this uh tentative islands um
should be embedded in something and
therefore we should intentionally say
there's a blanket or a encapsulation
which can be seen as like a channel like
a two-dimensional surface which acts as
a channel
but each channel mathematically has a
bottom
a limit so what if what happens when
that island that c space um generates
more entropy
than can actually be expressed through
the interface with wichita's relation to
the greater world
right um and so i'm basically trying to
indicate
um in part adam's suggestions

and intro and elaborate what i meant
with the difference between virtual
machines because
normal virtual machines aren't in this
mutually contextualized relation
not at the level of the physics as such
um and
because it's purely in the virtual
whereas here the physics
themselves are co-opted in that mutually
contextualized relation
okay sir that's enough for now yeah all
good thank you marco
so everyone thanks so much for
participating
if anyone wants to raise their hand and
give a final
just little tweet length
recap but what i'll say here is
organizer
is our next conversation live is going
to be
on february 23rd and we wrote
down a bunch of questions for next time
and there was also things that were
asked in the chat like about the
difference between quantum and classical
computers and simulation so i know that
we're going to have so many fun
directions to go to in our next
synchronous and asynchronous
conversations and communications through
the interface so
if anyone wants to give a final thought
um wanja
thanks so much for coming on it really
it was um it's a it's a special
paper it was a lot of fun to work

through enlivened our
lab and our collective thought processes
and
um yep so if anyone else wants to give
any final thoughts
um steven and then anyone else raises
their hand just for the last short
thoughts
steven unmute
just one just wanted to say thank you
very much and uh i know there's a lot of
stuff
has come at you so you you've managed to
field a huge amount of questions but i
uh
yeah i'd be really curious to maybe see
your thoughts on how this
this work can pan out in the world in
terms of
people's practices and ways of thinking
about
other research and how this kind of way
of understanding consciousness
could you know play out possibly in
because a lot of work in psychology is
using
brain neural correlates to justify what
they think people are thinking you know
so
it'd be interesting maybe next week to
think about how
how does some of these ideas have
implications and application
which uh i think they do so thank you
thanks steven
adam and then anyone else with a raised
hand
um yeah this i mean fantastic as usual

and this is uh
my it's the most fascinating thing there
is and i also think though it's
extremely important we get this right so
like marco is just discussing i'm
organizing we're talking about islands
of awareness
like how quickly do we move forward with
this research program
which could potentially you know help
countless people but we don't want to
inadvertently
create like black nourish scenarios so
if we can place
a very low prior unconsciousness for
organoids
due to not having the quite the proper
cybernetic
grounding contextualization and betting
that would be great like similarly like
with artificial intelligence we might
eventually uh create systems where
it is in question and so then we're
dealing with um
you know are these ethical beings on our
own are we like doing mind crime against
these beings or
like both avoiding negative outcomes and
then maybe
wanting there to be positive outcomes in
terms of creating
beings who are conscious and are of
value in their own right as subjects who
experience and there's something that
it's like to do them so they should be
part of our moral communities
all right these are maybe you know it's
it's not we're not there yet but

we might get there eventually and these
might be the questions that are the key
questions for
what really matters for those events
here yep and what's on the table adam as
you often highlight is everything from
who we include in our moral community
false positives and false negatives are
quite
vitally severe in this issue um
so looks like that's all the raised
hands
16.1 thanks everyone for coming this is
really a great discussion
and look forward to continuing this
discussion in 16.2 and beyond thanks